

Franklin Kuo

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EDUCATION

University of California, Berkeley

B.S. Mechanical Engineering

Berkeley, CA

Expected May 2029

EXPERIENCE

RSE Corporation

May 2026 – Present

Engineering Intern

- Ranked accident risk across all 43 Caltrain at-grade crossings to prioritize capital safety spending, by leading data analysis for the Grade Crossing Hazard Analysis.
- Enabled data-driven risk assessment via daily exposure metrics, by building Excel datasets tracking 24-hour incident events, vehicle, and cyclist volumes.
- Delivered final LA Metro Southeast Gateway Line cross-sections, by importing and cross-checking 100+ xref files against the shared ACC platform and building sections from the final CAD drawings.
- Cut hours of repetitive team work, by developing Python tools with Claude to inventory project files, consolidate incident logs into one master CSV, and auto-generate 43 hourly-volume chart sheets.

UC Berkeley Solar Vehicle Team (CalSol)

Jan 2026 – June 2026

Chassis Engineer, Occupant Cell Team

- Prevented wasted carbon fiber and excess vehicle weight on the 11th-gen “Excalibur” monocoque chassis, by catching and correcting a 3-degree CAD-to-manufacturing misalignment across the entire rear half of the occupant cell.
- Verified occupant-cell structural integrity under racing loads against stress safety factors, deflection limits, and American Solar Challenge roll-cage requirements, by running finite element analysis in ANSYS.
- Optimized occupant-cell geometry for a lightweight, high-strength carbon-fiber shell, by designing and revising components in SolidWorks against mass and aerodynamic targets.

Space Enterprise at Berkeley (SEB)

Sep. 2026 – Present

Avionics Team

Bana Builders, Inc.

June 2024 – Aug 2024

Engineering & Operations Intern

- Improved scheduling and workflow efficiency across active job sites, by coordinating construction operations directly with engineering staff and the project superintendent.
- Documented field changes for long-term project accuracy, by interpreting and producing architectural and as-built drawings in Bluebeam.

PROJECTS

Rail Alignment Geometry Automation Tool | *Python*

2026

- Automated a task that takes an engineer 2 to 4 weeks by hand, by building a Python tool (with Claude) that turns raw survey points into finished track alignment with ordered tracks, best-fit lines, curves, and spirals.
- Achieved 99% of points within 0.05 ft of the best-fit line at 10-ft intervals, by engineering an iterative tolerance sweep that re-runs and compares candidate geometries to select the optimal fit.
- Streamlined handoff, by outputting an Excel review report, a drawing, and a LandXML file for direct Civil 3D import.

Folding Game Table Mechanism | *SolidWorks*

Summer 2026

- Met both packaging targets for a 4 ft × 8 ft, 120 lb folding table — 4 ft folded width to fit a car and 7 in folded thickness against 7.5 in of garage clearance — by modeling the mechanism in SolidWorks and verifying fold geometry before cutting material.
- Balanced fold clearance against deployed stability at the leg pivots, by relieving the leg corners enough to clear the tabletop through the fold while preserving flat bearing surface to seat the legs when standing.

TECHNICAL SKILLS

Software: SolidWorks, ANSYS, AutoCAD, Civil 3D, Bluebeam, MATLAB, Microsoft Excel

Programming: Python, Java

Currently Studying: Machine Learning & Computer Vision (Stanford CS231N)

Languages: English (native), Mandarin Chinese (proficient)